Cheap Signals, Costly Proof: The Reach and Limits of Award-Layer Screening in Cartel Enforcement

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Abstract

Cartel enforcement must allocate costly proof-producing effort before legal proof exists. This paper asks how agencies should govern cheap administrative screens before opening bid-level records. Using São Paulo's BEC procurement platform and CADE adjudications, we audit a minimal award-layer construct—persistent zero-win participation—against a reproducible adjudication-anchored cobidder label that never uses the screen itself. The audit identifies an over-crediting problem: when both screen and label load on participation volume, raw metrics overstate evidentiary value—in BEC, an award-layer AUC of 0.76 falls to chance once opportunity is held fixed. Raw reach is exposure, not conduct, and strict prospective ranking fails outside incumbent pools. We then organize follow-up as a cost–recall frontier: one operating point opens 12% of firms but 67% of bid rows, so firm-count savings overstate forensic cost. A second-platform stress test shows that the audit logic travels; the score does not. Liability remains in the richer record.

Keywords: cartel screening, public procurement, award-layer data, enforcement triage, incomplete observability, bid rigging

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1. Introduction

Cartel enforcement begins long before legal proof exists, and its first problem is not adjudication but allocation. An agency deciding where to look rarely starts with

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the communications, bid histories, leniency testimony, or tender-level conduct evidence that would support liability. It starts with something thinner: administrative records, complaints, procurement anomalies, the ordinary residue of a market that some firms inhabit more than others. Which of these leads is worth the cost of pursuing is the question that precedes proof. The Becker–Stigler tradition treats investigation and sanctions as costly instruments whose intensity must be chosen optimally (Becker, 1968; Stigler, 1970); the margin we study sits one step earlier, where a resource-constrained authority must sequence costly evidence acquisition before it knows whom to charge. This is the enforcement-side counterpart to the procurement-governance problems in Decarolis et al. (2020): there a buyer must run a tender well, here an enforcer must decide where to spend forensic capacity once the tenders are run.

In procurement this margin has a sharp economic shape. The empirical cartel-screening literature usually works from the bid layer—the submitted amounts, ranks, dispersion, histories, and sometimes bidder networks that reveal how firms actually competed. Classic screens contrast bidding under competition and collusion (Porter and Zona, 1993; Bajari and Ye, 2003); variance and descriptive screens formalize suspicious bid distributions (Abrantes-Metz et al., 2006; Harrington, 2008) and field-test them in procurement (Imhof et al., 2018; Imhof, 2019); later work extends the approach through bidder groups (Conley and Decarolis, 2016), machine learning (Huber and Imhof, 2019; Wallimann et al., 2023), and robust screen design (Chassang et al., 2022). These objects sit close to the evidentiary stage, yet they are rarely cheap at scale. Many procurement systems make contract-award records routinely visible—who participated, who won, item codes, negotiated prices—while keeping per-bidder bid files in operational systems that must be recovered, cleaned, and inspected case by case. The award layer is cheap, broad, and legally thin; the bid layer is rich and costly to open. That asymmetry is the terrain of this paper. Cheap red flags of exactly this kind—single-bidding and repeat-participation indicators computed from award data—are already relied on by multilateral institutions and audit bodies precisely because bid files are expensive to recover (Fazekas and Kocsis, 2020; OECD, 2022); the zero-win construct we examine is the cartel-facing member of that family, a statistic an authority can run before committing to any forensic recovery.

Cheap screens of this kind are neither evidence nor noise. They are organizational inputs, and their value depends on whether an agency can separate behavioral information from exposure arithmetic before committing scarce forensic capacity. The contribution of this paper is therefore not a new cartel detector. It is an audit architecture for deciding when an administrative red flag should trigger costly bid-layer follow-up, when it should be read as ordinary opportunity exposure, and when it should be set aside.

We build it on São Paulo's BEC procurement platform, which covers 1,654,401 tender-items from 2009 to 2019 and records both layers—a routine award layer and a richer bid layer recoverable through the LANCES export—while CADE procurement-cartel adjudications supply external legal anchors. The combination lets us study a stage bid-distribution screens take as given: when an agency sees award records but has not yet paid to recover the evidence forensic evaluation requires.

The award-layer statistic is deliberately plain. Among firms with no wins, we rank participation intensity by log(1+tenders_count) and implement a transparent binary rule that flags *frequent losers*. The logic is monotone, not classificatory: if coordinated bidders use loser-side participation to manufacture the appearance of competition while ceding the win, persistent zero-win participation can rank loser-side exposure under one legible behavioral condition. The flag identifies no firm type and assigns no cartel membership; it allocates forensic attention, leaving the richer bid layer to judge tender-level conduct and agreement.

That scope dictates the validation target. Direct CADE defendants are legal anchors, yet they are not the natural object of a zero-win loser-side screen. Procurement cartels routinely allocate winning roles, rotate designated winners, and station other members on the losing side of the tender (Pesendorfer, 2000; Asker, 2010; Marshall and Marx, 2012), arrangements that large-scale procurement work shows surfacing in administrative bidding records (Kawai and Nakabayashi, 2022). We therefore validate the ranking against reproducible adjudication-anchored exposure rather than cartel membership. The target is the set of 651 unique always-loser firms that share at least one tender-item with a BEC-active direct CADE defendant, the defendants themselves excluded. The label is rebuilt end-to-end from the current pipeline, and—a point on which every subsequent comparison depends—the frequent-loser flag never enters its construction. The defendants anchor legally adjudicated environments; the always-loser cobidders are the loser-side footprint the award layer can plausibly rank before bid-level proof is recovered.

Three threats could each, on its own, hollow out the ranking. The first is opportunity arithmetic: high-volume zero-win firms mechanically meet more defendants, so a raw co-bidding association may track participation volume rather than any behavioral signal. The second is timing: a ranking built from contemporaneous or later participation cannot order forensic priority prospectively. The third is case concentration, where one adjudicated case manufactures apparent platform-wide reach. We hold the ranking against all three—opportunity-cell adjustment, strict rolling-origin timing, and leave-one-case-out tests—alongside a bid-layer benchmark; Section 4 and Appendix C report the full battery and the exact exposure, leakage, and timing construction.

What the audit returns is sobering and instructive. Read raw, the ranking does concentrate the 651 adjudication-anchored cobidders inside the always-loser stratum (ROC-AUC 0.761). The decomposition then shows how little of that is conduct. Most is procurement opportunity: firms that participate more, in the same environments as adjudicated defendants, meet them more often. Hold opportunity fixed and the residual ordering collapses—AUC 0.471 within matched-opportunity strata, at chance; a +0.010 increment over an exposure-only model (DeLong $p = 0.013$) too small to reorder a forensic queue; matched-stratum permutation and enrichment tests that cannot reject the opportunity-only null. An opportunity ranking that never sees the score already reaches 0.713 against the raw 0.761. Strict prospective ranking then fails outside the incumbent pool—on the full candidate universe precision@500 is exactly zero—and the reach is case-sensitive: a single adjudicated case supplies 32.0% of the positives. The ranking is also silent about direct CADE defendants, exactly as its loser-side scope predicts.

Where the design earns its keep is in cost accounting and forensic sequencing. On the same adjudication-anchored target the award-layer ranking is comparable to a transparent bid-moment random-forest benchmark in the spirit of Imhof–Wallimann screens, and the two combine only conditionally: a joint model improves precision under random cross-validation but falls below the award-only ranking under case-grouped folds. Used as a gatekeeper, the award layer traces a cost–recall frontier—across survivor-pool sizes it retains much of the recall of full bid-layer scoring while shrinking the bid-recovery footprint. The reduction is real, but must be read against its denominator: large in firms opened, far smaller in tender-items and bid rows, because the survivors are high-participation firms. No single cutoff is optimal; the frontier itself is the enforcement-design object.

The paper makes three contributions; the first two are analytical objects the screening literature has not written down.

The first, and the one most likely to travel beyond procurement, is that the over-crediting of cheap screens is itself an estimable object, not a caveat. A contact-defined label and a participation-based score both load on the same volume, so the positive class is just the size-biased participation distribution. Validating such a screen against adjudicated cases without adjusting for opportunity does not merely risk inflation; it inflates by an amount whose signs we pin down—a size-bias gap that grows with the dispersion of participation volume and shrinks with the adjudicated base rate (Proposition 1), with magnitude read from a synthetic surface anchored at one empirical point per platform. The practitioner deliverable is a leading-order sufficient statistic—the coefficient of variation of participation counts, computable from award data before any bid file is opened—that bounds how much of a raw screening number is mechanical. The two platforms are then two points on one surface: the federal system's rarer target is exactly why its opportunity-only model edges past the raw score there, as the characterization predicts.

The second is organizational, and it carries the law-and-economics payload. We model the award-to-bid recovery decision as the enforcer's evidence-acquisition problem: an agency with per-record forensic cost c and per-case recovery payoff V descends a cheap ranking until the marginal case recovered per forensic dollar meets the cost–value ratio c/V (Proposition 5). The cost–recall frontier we report is the empirical image of that tangency, and "there is no optimal cutoff" follows mechanically: the optimum is budget-dependent, so the designer reports the locus of those optima—the frontier itself—each operating point, $K_1 = 2{,}000$ included, the tangency for one implied c/V.

Procurement-cartel screening thus stops being a classification exercise and becomes a sequencing problem in the optimal-enforcement tradition: how a constrained authority orders costly evidence acquisition when suspicion is cheap and proof is dear (Becker, 1968; Polinsky and Shavell, 2000; Townsend, 1979; Mookherjee and Png, 1989). Bid-distribution screens evaluate suspicious bidding once the bid-level data are in hand; we study the prior stage, where the frontier, not a detector-versus-detector horse race, is what matters.

The third is the reproducible, non-circular audit protocol that produces both objects. Its components are individually familiar—an adjudication-anchored label that never uses the flag, opportunity-cell adjustment, rolling-origin timing, leave-one-case-out tests, a bid-layer benchmark, and cost–recall accounting—but their sequencing for the award-to-bid decision is, to our knowledge, new. The estimated ranking is not a portable cartel score—the evidence denies that object—but the stopping rule, the inflation characterization, and the protocol itself are. We test that portability directly: re-running the identical battery on the federal ComprasNet platform (Section 5), with partially overlapping CADE anchors and no bid microdata, reproduces the first-stage deflation and places the second platform where the synthetic surface predicts. Throughout, price evidence enters only as scope, never as damages, and because a published cutoff invites gaming the design favors refreshed thresholds and continuous ranks over any fixed one.

The institutional lesson is plain. A legal system that waits for full proof before ranking investigative priorities will spend scarce forensic capacity badly; one that treats cheap suspicion as proof will overreach. The workable middle ground is a disciplined triage rule—cheap enough to run before proof exists, informative enough to order forensic attention within a known reach, and limited enough to leave liability in the richer record where it belongs. Knowing where such a rule stops ranking is as much a part of sound enforcement as knowing where it begins.

2. Costly Proof and Observable Awards

Our empirical object is an institutional sequence: an agency first observes a thin administrative record, then decides whether to spend resources recovering richer bid-level evidence, and only later can that evidence contribute to legal proof. BEC procurement supplies the sequence cleanly—an award layer, a bid layer, and the adjudication-anchored labels we use to validate a loser-side ranking. The legal predicate fixes the screen's role. Proof of a procurement cartel requires evidence of coordinated conduct—communications, allocation rules, bid rotation, cover bidding, or tender-level patterns inconsistent with independent rivalry. Brazilian law treats bid rigging as an infraction under Lei 12.529/2011, and comparable frameworks likewise demand evidence beyond suspicious administrative patterns (Hovenkamp, 2005; Baker, 2003; Wils, 2007, 2016). Award records contain none of those predicates, so an award-layer screen produces *forensic priority*, not an accusation: it orders where costly bid recovery should start, while liability stays in the bid layer.

2.1. Award and Bid Layers in BEC

São Paulo's BEC platform preserves both layers. The sample contains 1,654,401 tender-items from 2009 to 2019, dominated by two modalities: *Convite*, a sealed-bid invitation procedure under Lei 8.666/1993 with a statutory three-bidder quorum and a low value cap; and *Pregão*, a reverse electronic auction under Lei 10.520/2002 with real-time offers and no minimum-bidder rule.[2] Both generate a routine award record before any case-specific bid recovery occurs, even as auction format, bureaucratic incentives, discretion, and rules shape bidding and outcomes (Athey et al., 2011; Decarolis, 2014; Coviello and Gagliarducci, 2017; Decarolis et al., 2020, 2025).

The *award layer* is that routine administrative envelope, recording participant and winner identity, item code, negotiated price, procuring unit, year, and modality—who appeared, who won, what was bought, and where a firm repeatedly participated without winning. It reveals none of each participant's bids, their timing, revisions, or within-tender dispersion: cheap enough for broad monitoring but too thin to prove coordination.

The *bid layer* is the forensic-recoverable layer. BEC preserves per-bidder offers, timestamps, and bid-revision sequences in the LANCES export, which support bid-distribution forensics—within-tender dispersion, skewness, spread, distance from the

[2]A firm's participation is counted once per tender-item it enters. In Pregão, a bidder may revise its offer repeatedly within a single tender-item; these iterative offers do not raise the participation count T_i, defined at the tender-item level.

winning bid, and related bid moments inspired by Imhof–Wallimann-style screens (Imhof et al., 2018; Imhof, 2019) and the machine-learning variants that combine them at scale (Huber and Imhof, 2019; Wallimann et al., 2023). But they must first be recovered, cleaned, linked, and inspected, at a cost and timing no award-record query incurs—feasible for case work and validation, not the natural first object for system-wide triage.

The closest antecedent to a loser-side collusion marker also lives in the bid layer. Seibel and Škoda (2026) derive, from a model of partial cartels under bidder preselection, a theory-grounded exclusion marker and show on Slovak procurement microdata that it discriminates suspected colluders. Our object is deliberately thinner—an atheoretic loser-side participation rank read off the award layer, not a structurally derived marker estimated on bids—and the comparison sharpens the opposite point. A loser-side signal that carries content in rich bid data has a cheap award-layer analogue, and we show that the analogue largely reflects procurement opportunity rather than conduct (Section 4). The deflation under opportunity adjustment marks how far down the information ladder a loser-side construct can be pushed before it stops measuring coordination and starts measuring exposure.

2.2. CADE Anchors and Adjudication-Anchored Exposure Labels

CADE adjudications supply legal anchors, not the screen's labels. The linked BEC portfolio contains 12 adjudicated procurement-cartel cases with 65 firm-defendants, of which 47 are active in BEC during the sample window. Final decisions span 2015–2025, with long process-to-ruling lags. That timing is central: whatever value a cheap ranking has lies in moving investigative attention before the completed adjudication record exists—a premise the paper audits rather than assumes.

The record creates two objects with different roles. Direct CADE defendants—firms named in adjudicated cases and matched to BEC participant records—are the legal anchors. They locate environments where legal process established cartel conduct, but they are not the natural target of a zero-win statistic: defendants may occupy winner-side, allocation, or rotating roles, so a screen built from repeated losing should not be judged as a membership classifier over them.

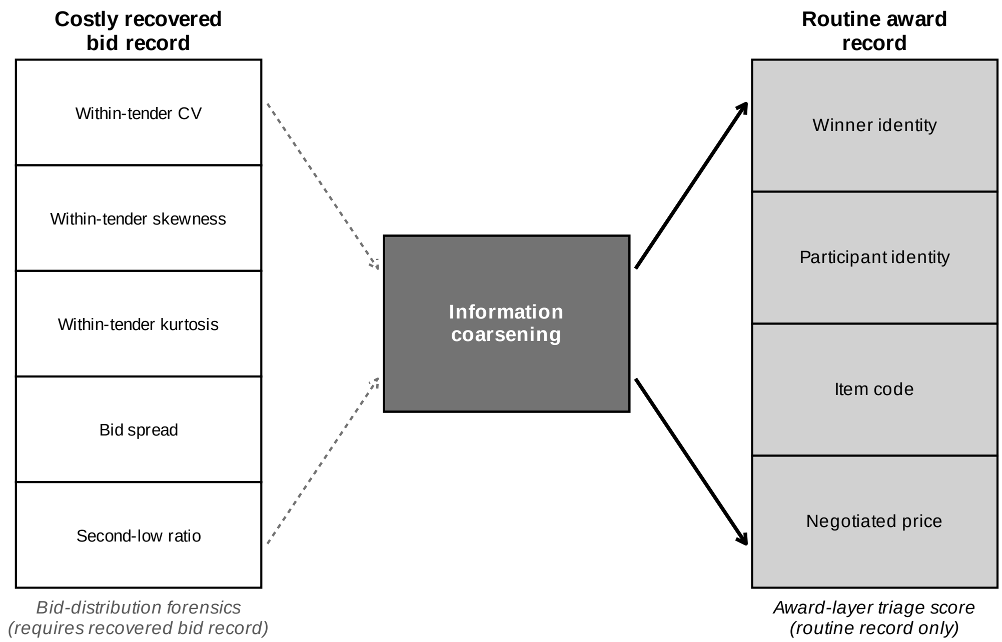

Figure 1: Information layers in procurement-cartel enforcement—an information-cost diagram, not a detector horse race. Bid-distribution forensics needs per-bidder amounts and within-tender bid moments; the award-layer screen uses participant and winner identity, item code, and negotiated price. The bid layer is not absent but costly to recover from the LANCES export. The question is whether the cheap award layer can rank where the costly bid layer should be opened first. A pattern in award records does not imply legal proof of coordination.

Alt text: Two vertical columns of stacked boxes represent two information layers. The left column lists costly recovered bid-record features (within-tender bid moments such as dispersion, skewness, spread, and second-low ratio). The right column lists routine award-record fields (winner identity, participant identity, item code, negotiated price). A central gate links them: the thinner award layer is available earlier and more cheaply and is used to rank where the costly bid layer should be recovered first.

The validation target is narrower. A cobidder is a unique always-loser firm that shares at least one BEC tender-item with a BEC-active direct CADE defendant, yielding 651 always-loser cobidders; of the 651, 341 are frequent losers and 310 are not, so the flag plays no part in constructing the label. Cobidder status is not legal membership or a latent truth about agreement but an adjudication-anchored exposure label: the firm is observed on the losing side near a legally established cartel anchor, fit for evaluating forensic priority, not liability. The question is whether persistent zero-win participation concentrates the loser-side firms most plausibly worth bid-layer follow-up around adjudicated cartel environments—which Section 4 disciplines against opportunity, timing, and case concentration.

Table 1 fixes the roles used throughout: the outer BEC population, the always-loser stratum the award layer can rank, the frequent-loser operational screen, the direct-defendant legal anchors, and the cobidder validation target.

The rest of the paper keeps these objects separate: the screen operates inside the always-loser stratum, validation targets cobidder adjacency, direct defendants anchor the environments without defining the screen's success, and bid-layer forensics remains the proof-producing stage. The separation lets us evaluate a legally limited triage rule without turning suspicion into adjudication.

3. Award-Layer Triage

The award layer observes outcomes before it observes conduct. A losing bid may be sincere, poorly priced, capacity constrained, exploratory, strategic, or coordinated, so the screen uses only what the routine record measures at the triage stage: which firms appeared, which won, and which kept appearing without ever winning. The object is a ranking for forensic priority, not proof of conduct or a classification of cartel membership. What follows sets out that ranking, the operational queue it yields, and the implications that constrain its reading.

Table 1: Data Layers, Populations, and Legal-Economic Roles

Object	Size / layer	Role in the analysis	Legal interpretation
Award layer	Routine record	Participants, winners, item codes, prices, procuring units, years, and modalities; the cheap triage layer.	Administrative record; not proof.
Bid layer	LANCES export	Per-bidder offers, timestamps, revisions, and bid moments; the costly forensic layer.	Forensic record; can support bid-level analysis.
All BEC participants	41,444 firms	Outer firm universe for award-record construction.	Outer universe.
Always-losers	16,843 firms	Zero-win stratum in which the award-layer screen ranks loser-side exposure.	Screening stratum.
Frequent losers	2,735 firms	Operational first-stage flag within always-losers; not a legal category.	Operational flag; not a legal category.
Direct CADE defendants in BEC	47 firms	Legal anchors from adjudicated procurement-cartel cases.	Legal anchor.
Always-loser cobidders	651 firms	Always-losers sharing a tender-item with direct defendants; main validation target.	Exposure label; not membership.

Notes: Direct CADE defendants are legal anchors. Always-loser cobidders are adjudication-anchored exposure labels. Frequent losers are an operational screen, not a legal category. The categories separate legal membership from screening exposure. Benchmark-specific cobidder counts and the funnel reconciliation appear in Table 2.

3.1. Persistent Zero-Win Participation

Let T_i denote the number of BEC tender-items in which firm i participates (T_i counts tender-item participations, so iterative Pregão offers within one tender-item do not raise it), and let W_i denote the number it wins. The screen operates inside the zero-win stratum, $\{i : W_i = 0\}$, which contains 16,843 firms. Within it, the continuous score is

$$s_i = \log(1 + T_i).$$

The log stabilizes the scale without changing the ordering. The economic object is the rank: among firms the award layer observes only as losers, which ones appear persistently enough to warrant bid-layer follow-up?

The maintained condition is monotonic, not classificatory. If cartel-deployed losing participants are present, they should appear more persistently in the zero-win stratum than ordinary losing firms, because such a role pays only through repeated deployment in cartel-targeted tenders. Appendix B formalizes this as a monotone-likelihood-ratio ranking result; the main text uses it more modestly. Persistent zero-win participation is a cheap statistic that ranks loser-side exposure under a transparent behavioral assumption and directs attention to the bid layer, where conduct can be evaluated—ordering firms for forensic priority without assigning a type.

The zero-win restriction is a scope restriction: most firms never win for the ordinary competitive reasons noted above, and the screen's content comes from persistence inside that heterogeneous loser-side set. Appendix B adds the dynamic margin: ordinary zero-win firms face participation costs and may exit after repeated losses, while a losing role that carries strategic value or compensation can sustain participation. A platform-survival audit indeed finds that frequent-loser cobidders stay active markedly longer than other frequent losers—persistence as a triage signal, not a legal category.

3.2. From Ranking to an Operational Queue

The continuous score is the central object because agencies can translate a rank into different investigative protocols: inspect the top k firms under fixed capacity, refresh the rank each quarter and examine bunching below old cutoffs, or request LANCES bid files for a survivor pool before applying bid-distribution forensics. All preserve the same sequencing logic: award records first, costly bid recovery second.

For an auditable binary implementation, we also report a frequent-loser flag marking zero-win firms whose participation count exceeds the median plus 1.5 times the interquartile range of the always-loser participation distribution. In BEC, the statistical cutoff is 13.5, implemented as $T_i \geq 14$, which flags 2,735 firms, or 16.2% of the always-loser stratum. The cutoff is fixed from the participation distribution alone, before any CADE label enters the analysis.

The $T_i \geq 14$ rule is an auditable administrative implementation of the rank, not a structural or legal threshold; the paper rests on the continuous ranking, not the number 14. Two analysts applying the same rule to the same records obtain the same first-stage list, but above the cutoff does not mean cartel member and below it does not mean clean. Because a published static threshold can be gamed, the design treats the binary flag as one implementation among several—continuous ranks, top-k queues, refreshed thresholds, budget-calibrated survivor pools—a point Section 8.1 revisits.

3.3. Testable Implications

The ranking can organize incumbent-firm triage only if it survives the threats implied by its limited information. The core implication is the validation claim: persistent zero-win participation should rank adjudication-anchored loser-side cobidders within the always-loser stratum, beyond what mechanical exposure—high participation volume and operation in the same environments as CADE defendants—would produce under ordinary competition. The rest follow: scope (the statistic should perform worse against direct defendants than cobidders, ranking a different role), the economic content of the cobidder target, non-redundancy relative to bid-distribution forensics, gatekeeping value at affordable budgets, and vulnerability of the published binary threshold to strategic adaptation. Section 4 and the appendix evaluate each against the threats that would weaken it, assessing the screen at the stage it is designed for: allocating costly forensic attention before liability evidence is assembled. The BEC cutoff is an implementation detail; the portable object is the staged design itself.

4. Validating the Loser-Side Ranking

What follows validates the award-layer ranking, not the legal status of ranked firms. The validation target is the one defined in Section 2.2: always-loser firms observed alongside direct CADE defendants in adjudicated procurement-cartel environments. These cobidders carry an adjudication-anchored exposure label, not cartel membership. A useful ranking must clear more than a high AUC. The dominant threat is exposure: firms that participate often, or operate in the same environments as defendants, have more opportunities to appear near those anchors even under ordinary competition, so a screen earns credit only by concentrating loser-side firms beyond what participation volume and opportunity alone imply (Porter and Zona, 1993; Bajari and Ye, 2003; Conley and Decarolis, 2016; Kawai and Nakabayashi, 2022). Timing and legal overreach are the two secondary threats: the screen should rank before the target information is used, and it should not be read as a generic cartel-membership detector. The exposure threat is the award-layer counterpart of a problem Kawai et al. (2023) make precise in the bid layer, where winner-and-loser allocation patterns can reflect cost differences or conduct and separating the two requires identifying variation. We do not observe bids, so the opportunity-versus-conduct decomposition that follows is the award-layer analogue of that test, asking what the loser-side ranking adds once procurement opportunity is held fixed. Around that central exercise, timing and case-composition checks and a direct-defendant scope bound complete the validation.

4.1. Label Construction and Sample Reconciliation

The CADE legal portfolio comprises 12 adjudicated procurement-cartel cases with 65 firm-defendants. Matching defendants to BEC records by exact 14-digit CNPJ equality, with no fuzzy or manual step, leaves 47 active in BEC as the direct-defendant set, of which 41 appear in the firm–tender map and anchor the label funnel; their 52,013 defendant tender-items locate the adjudicated environments. The main label collects unique always-loser firms (win rate = 0 over 2009–2019) that share at least one BEC tender-item with those anchors, excluding the defendants themselves. Crucially, the frequent-loser flag never enters the label: it is the award-layer score being *evaluated* against the label. The match yields 651 always-loser cobidders, 341 frequent losers and 310 not—the label straddles both sides of the screen's threshold, so the validation cannot be circular in the score. A conservative benchmark restricts the portfolio to the 4 cases judged on or before 2020 under the same definition (208 cobidders from 19 defendants), a strict subset of the main label (Table 2). Positive counts differ across later tables only because later exercises impose common-support, timing, or bid-feature restrictions; Table A.3 in the appendix reconciles each denominator.

Table 2: Label-Construction Funnel and Benchmark Reconciliation

Sample / benchmark	Cases	Legal def.	BEC-active def.	AL cobid.	FL among	Cobidder def.	Role
Full CADE portfolio	12	65	47	651	341	shared tender-item	reference
Main validation target	12	65	41	651	341	shared tender-item	primary
Conservative pre-2020	4	–	19	208	107	shared tender-item (same)	early-cases subset

Notes: A cobidder is an adjudication-anchored exposure label, not membership. The main label is a unique always-loser firm (win rate = 0 over 2009–2019) sharing at least one BEC tender-item with a BEC-active direct CADE defendant; direct defendants are excluded; the frequent-loser flag is not used to construct the label. "FL among" is the descriptive composition of positives, not a label restriction. The 47 cited active defendants correspond to 48 distinct crossmatch CNPJ, of which 41 are present in the firm–tender map. Counts are unique firms.

A baseline question precedes any discipline: do firms flagged by the award-layer rule cluster among always-loser firms near adjudicated anchors? They do, and markedly. A frequent loser is a cobidder with probability 12.5% (341/2,735), against 2.2% for non-frequent-loser always-losers (310/14,108)—an enrichment near 5.7×. On the full always-loser pool the continuous score ranks the label at ROC-AUC 0.761 (PR-AUC 0.143; precision@500 0.216, lift 5.6×), the binary flag at 0.688. Persistent zero-win participation is plainly not random with respect to adjudication-anchored loser-side exposure. But participation volume and shared environments mechanically manufacture proximity, so whether the pattern reflects anything beyond arithmetic is the open question—and, as the next subsection shows, very little of it survives once opportunity is held comparable.

4.2. Opportunity-Adjusted Validation

The threat here is mechanical: a firm with high participation T_i co-appears with defendants more often by construction, and a firm sharing item, buyer, year, and modality cells is more exposed even under ordinary competition. We measure opportunity directly. For each always-loser firm we count observed contact with direct defendants, O_i, and expected contact under random matching, $E_i = \sum_g n_{ig}\, p_g$, where n_{ig} is participation in cell g and p_g its defendant-density. We build E_i at three cell granularities and restrict to common support; the exposed subsample retains $n = 6{,}040$ always-loser firms containing all 651 positives (construction in Appendix C.1). Holding E_i fixed, we ask whether the score reproduces the label better than exposure alone, adds anything net of it, and ranks within equal-opportunity strata.

The answer cuts the raw performance down on every margin, and the exposure benchmarks must be read in tiers because contact-derived quantities partially encode a contact-defined label. The mechanical-extreme benchmarks—ranking by observed contact O_i, its plug-in E_i, and the combined logits, all near or above ROC-AUC 0.9—encode the label (a cobidder *is* a firm with $O_i > 0$) and are reported only to expose the inflation (Table 3). The decisive read is the label-blind benchmark: recomputing each cell's defendant density from *non-cobidder* rows only, so no eventual positive contributes to any firm's expected contact, leaves an opportunity ranking at just 0.553 (the unpurged leave-one-out variant is 0.855; Appendix C). Opportunity structure purged of the label explains little on its own.

What the tiers establish is sharper than "opportunity is the better model," and the mechanism is exact. The award-layer score is $\log(1 + T_i)$, monotone in participation volume T_i. The cobidder label is contact-defined: a positive is an always-loser sharing at least one tender-item with a defendant, so the probability of being labelled also rises in T_i—more tender-items, more chances to contact a defendant. Score and label are both monotone transforms of the same volume primitive. A firm need not coordinate to be ranked high and labelled positive; it need only participate often. This makes the positive class the size-biased participation distribution, so the raw ROC-AUC is just $\Pr(T_i > T_j)$ with T_i size-biased relative to a random negative T_j—an ordering that exists before any conduct enters. Raw validation over-credits the screen because it credits volume geometry to it. The over-crediting is thus a predictable bias, not a quirk of this dataset, and its magnitude turns on only two quantities: the dispersion of participation volume and the adjudicated base rate. Writing the over-crediting gap as $\Delta = \mathrm{AUC}_{\text{raw}} - \mathrm{AUC}_{\text{opp-adj}}$, its signs follow from the size-biasing mechanism alone.

Proposition 1 (Signs of the over-crediting bias). *Let $\Delta = \mathrm{AUC}_{\text{raw}} - \mathrm{AUC}_{\text{opp-adj}}$, where the raw award-layer area is the size-biased ordering $\mathrm{AUC}_{\text{raw}} = \Pr(T_i > T_j)$ with the positive T_i size-biased relative to a random negative T_j, and $\mathrm{AUC}_{\text{opp-adj}}$ holds the participation-volume channel fixed. In the rare-target regime:*

(a) Δ is increasing in the dispersion of the participation-volume distribution F_T; for T in a scale family it is increasing in the coefficient of variation $\mathrm{CV}(T)$, with $\Delta \to 0$ as $\mathrm{CV}(T) \to 0$ (a degenerate T leaves nothing to rank);

(b) Δ is decreasing in the adjudicated base rate: as the base rate rises the positive class ceases to be as strongly size-biased and $\mathrm{AUC}_{\text{raw}}$ declines toward the genuine-signal floor.

Proof in Appendix C. The signs are analytic; the magnitude is not closed-form once one leaves the rare-target linearization and uses the empirical heavy-tailed F_T, so we characterize it by a synthetic surface (anchored at one empirical point per platform), not an estimated curve: Figure C.1 (Appendix C) maps Δ over a grid in $\mathrm{CV}(T)$ and the base rate, with BEC overlaid at its measured coordinate. The takeaway is portable: the coefficient of variation of T in the candidate pool is a *leading-order sufficient statistic* (within the scale-family approximation) for how inflated an uncorrected award-screen AUC is likely to be, computable from award data alone before a single bid file is opened. It diagnoses the size of the bias; it does not rescue the screen.

Conditional on opportunity, the residual ordering is marginal and fragile, though the within-stratum design passes an audit of its power to detect. Adding the score to the exposure-only model raises AUC by 0.010 (DeLong $p = 0.013$; DeLong et al., 1988). Read magnitude-first, a one-percentage-point gain sits far below any margin that would reorder a forensic queue, even though it is the only test in the opportunity-adjusted battery that is both positive and conventionally significant (and the only one to survive Benjamini–Hochberg). *Within* opportunity strata—firms compared only against others with comparable expected contact—the continuous score ranks the label at AUC 0.471 and the binary flag at 0.507, both at chance. The null is falsifiable, not built in: in the same strata a positive-control score (O_i itself) ranks the label at 0.953, so the design recovers a residual when one exists. Discretizing the strata pushes the within-stratum score AUC only to 0.493 in medium cells, rising toward 0.600 in the strictest cells only where support thins and the O_i control itself drops—low support, not a recovered residual. The remaining variants line up the same way: a coarsened-exact opportunity match leaves a negative flagged-versus-non-flagged gap; a label-blind stratification retains 0.722, but the placebo and non-CADE high-volume anchors of Section 6 reproduce that ordering, marking it as generic co-participation geometry, not anything cartel-specific (full granularity sweep, CEM match, and the excess-contact diagnostic in Appendix C).

Exposure-preserving permutation tests settle the question. Under a pure-exposure binomial null the observed PR-AUC 0.143 does not beat the null mean of 0.264 ($p = 1.00$); the pure-exposure PR-AUC in fact exceeds the raw score's. Shuffling cobidder labels only within matched exposure-by-participation strata—the design that isolates any residual signal—likewise fails to reject ($p = 0.127$), as does the frequent-loser enrichment within those strata ($p = 0.067$). These non-rejections are informative rather than underpowered: injecting synthetic residual signal shows the test is correctly sized and rejects with probability 0.97 once the true within-stratum AUC reaches 0.55 (1.00 at 0.60), bounding any surviving residual below the weakest ordering that could matter operationally. In rule-out terms, because the design would detect a within-stratum residual of AUC 0.55 with probability 0.97, we can bound any genuine residual ordering below that threshold—beneath the weakest signal that could reorder a forensic queue. Label breadth does not drive the result either: restricting positives to firms sharing at least two defendant tender-items leaves the within-stratum AUC at chance and the nested increment vanishing (both permutation distributions and the intensity-restricted breadth check in Appendix C).

The reading is therefore stronger than caution. Under a reproducible, non-circular label, holding opportunity fixed leaves the award-layer score without a residual behavioral ranking: the apparent reach of the routine record is opportunity arithmetic, so loser-side concentration should not stand alone as a prioritization device here—while the same decomposition, run on another platform, would show where, if anywhere, a cheap screen retains residual value. Timing, leakage, and case composition sharpen the same reading next.

Table 3: Opportunity-Adjusted Validation of the Award-Layer Ranking

Design	Score	N	Pos.	ROC-AUC	PR-AUC	Prec@500	Rec@500	95% CI / p
Raw score	log_tc	16,843	651	0.761	0.143	0.216	0.166	$[0.741, 0.780]$
Raw FL14	fl14	16,843	651	0.688	0.097	0.128	0.098	$[0.669, 0.707]$
Exposure-only benchmark	log E_i	6,040	651	0.713	0.300	0.372	0.286	$[0.691, 0.735]$
Within-opportunity stratum	log_tc | opp	6,040	651	0.471	—	—	—	pooled C (chance)
Within-opportunity stratum	fl14 | opp	6,040	651	0.507	—	—	—	pooled C (chance)
Nested increment over exposure	+log_tc	6,040	651	+0.010	—	—	—	DeLong $p = 0.013$
Score ranks excess contact	log_tc	16,843	475	0.764	0.095	0.150	0.158	$1[O_i - E_i > 0]$
Permutation B (pure exposure null)	log_tc	16,843	651	—	0.143	—	—	$p = 1.00$
Permutation C (matched strata)	log_tc	16,843	651	—	0.143	—	—	$p = 0.127$
Matched / stratified (CEM-opp)	log_tc	6,040	651	0.626	—	—	—	$\Delta = -0.017$

Notes: The validation target is adjudication-anchored exposure (cobidder status) among always-loser firms, not cartel membership; the label never uses the frequent-loser flag (341 of 651 positives are frequent losers, 310 are not). Expected contact $E_i = \sum_g n_{ig} p_g$ is built from procurement opportunity cells (item, buyer, year, modality); cell definitions and retained support are in Appendix C.1. PR-AUC is the primary metric because the target is rare (651/16,843). The exposure-only benchmark reproduces the label better than the raw score on the rare-target metric; within-stratum and matched designs are at or below chance; the matched-strata permutation does not reject. Combined exposure-plus-score logits (ROC-AUC near 0.99) are deliberately omitted because E_i mechanically encodes the label and their fit is not score evidence.

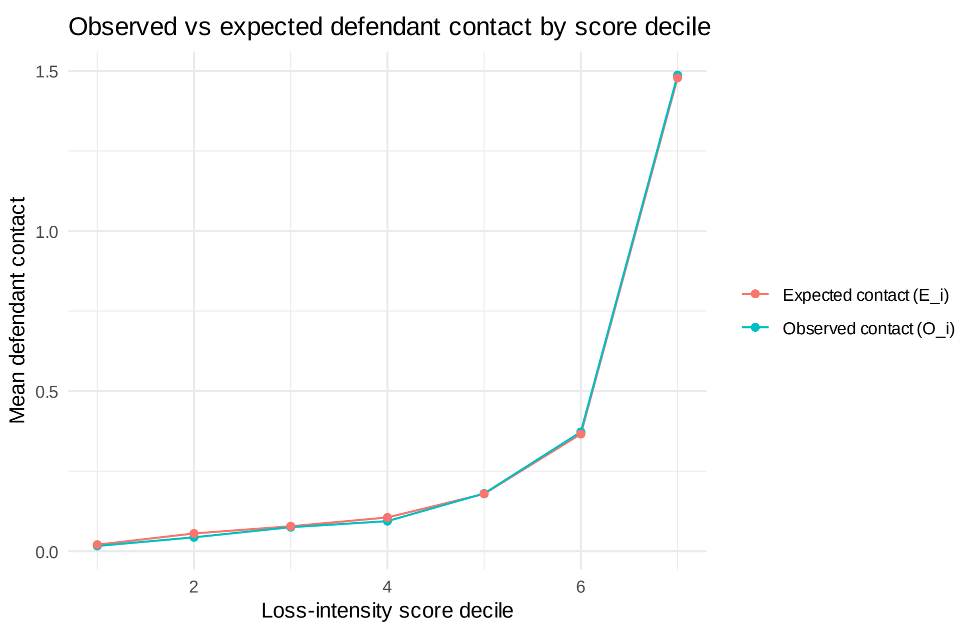

Figure 2: Opportunity-adjusted diagnostic: observed defendant contact O_i versus expected contact E_i under random matching, by award-layer score bin. Expected contact is built from procurement opportunity cells (item, buyer, year, modality); the comparison shows how much of the apparent score–contact relation survives the opportunity benchmark. *Alt text:* A binned plot with award-layer score bins on the horizontal axis and defendant-contact counts on the vertical axis. Two series are shown per bin, observed contact and expected contact under random matching; the two series track each other closely across score bins, indicating that expected contact under random matching accounts for most of the observed gradient.

4.3. Timing, Leakage, and Case Composition

Even without a robust residual ordering, an agency still needs to know whether the raw ranking can be applied prospectively. It cannot: the ranking is a retrospective, incumbent-pool diagnostic concentrated in a few adjudicated cases, and it does not support platform-wide deployment.

Every benchmark anchors its labels to CADE adjudications that close years after the bidding they describe, so even a high in-sample AUC would certify construct validity, not field performance. The strict-timing design freezes the score, always-loser status, and frequent-loser threshold to the 2009–2016 window and evaluates against 2017–2019 labels. On the realistic full candidate universe—every firm scorable at the screening date, entrants carried at score zero—discrimination collapses to chance or below: ROC-AUC 0.474, PR-AUC 0.014, and **precision@500 and recall@500 both** 0. A screen returning zero true positives in its top 500 cannot be deployed. Inside the training always-loser pool the frozen score retains ROC-AUC 0.684 (binary flag 0.646)—the incumbent-triage residue, an ordering among firms the agency already observes. The cause is structural: of the 651 positives, 153 (23.5%) are entrants with no pre-window history, so the frozen score cannot rank them; only 498 are rankable at all, and roughly a quarter of the universe is tied at score zero. A loser-side score built from past losing cannot order firms that have not yet lost. Two further checks confirm the collapse and close the residual leakage concern: a rolling-origin design returns zero precision@500 every test year from 2014 to 2019, and a fully label-frozen variant—freezing the pool, the score, *and* the label ingredients to 2009–2016—ranks new 2017–2019 defendant contact only as participation volume forecasting generic future contact with active firms, not a cartel-specific signal the stricter designs missed (both in Appendix C). The full-universe zeros remain the operative deployment result.

Case composition is the second structural fact. One case (trens/metros, CADE process 08700.004617/2013-41) supplies 32.0% of all positives and 45.4% of the top-500 true positives; dropping it cuts PR-AUC from 0.143 to 0.090 (−37%) and precision@500 from 0.216 to 0.132 while ROC barely moves—exactly why ROC is the wrong lead metric, being insensitive to the concentration the operational metrics expose. Equal case re-weighting pulls precision@500 to 0.043, and the signal also concentrates across environments (85.4% of positives sit in Pregão; the largest item-group holds 21.5%). A clustered randomization-inference test (item-group×year cells, $B = 1{,}000$) distinguishes the pooled ordering from a within-cell shuffle ($p = 0.001$) but not case-coverage breadth ($p = 0.103$): the raw ranking picks up concentrated exposure, not breadth. The leave-one-case-out distribution is in Appendix C, and Table 4 synthesizes the battery.

At most, then, this design supports incumbent-firm triage under retrospective validation—one input to where a forensic team looks first, explicitly not a prospective platform-wide screen. The estimated ranking is not a portable cartel score; the transferable object is the decomposition itself, taken up below.

Table 4: Timing and Case-Composition Synthesis

Design	Information set	N	Pos.	ROC-AUC	PR-AUC	Prec@500	Rec@500	Interpretation
Full-sample retrospective diagnostic	Full retrospective	16,843	651	0.761	0.143	0.216	0.166	Construct validity only; upper bound
Strict full test universe	Strict; entrants at score 0	41,444	651	0.474	0.014	0	0	Collapse: no prospective ranking
Strict rankable (incumbent)	Strict; prior-history firms	32,682	498	0.471	0.013	0	0	Entrant exclusion does not rescue
Strict training always-loser pool	Strict; train-window AL	21,819	651	0.684	0.077	0.124	0.095	Incumbent-triage residue
Rolling-origin (2014–2019)	Strict; full universe	per-year	381–651	0.45–0.50	≈ 0.012	0	0	Zero at the top every year
Leave-largest-case-out	Full retrospective	16,393	443	0.763	0.090	0.132	0.149	−37% PR-AUC
Leave-top-two-cases-out	Full retrospective	16,361	268	0.752	0.058	0.092	0.172	Thin residual tail
Clustered randomization inference	Label shuffle in cells	16,843	651	0.761	0.143	0.216	—	$p = 0.001$; coverage $p = 0.103$
Direct-defendant scope	Strict and full scores	—	47	0.49 (binary) / 0.66–0.70 (continuous)	—	—	—	Scope boundary (Section 4.4)

Notes: PR-AUC, precision@500, and recall@500 are the primary metrics because the target is rare; ROC-AUC is reported alongside but is insensitive to the case concentration the operational metrics expose. The full-sample retrospective row is a diagnostic *upper bound*, not a timing claim. In the strict and rolling-origin designs the score is frozen before the test window; later CADE adjudications are still used as labels but were not legally observable at the screening date, so these designs distinguish prospective scoring from retrospective adjudication. Cobidders are adjudication-anchored exposure labels, not cartel members. The strict full universe carries entrants at score zero; roughly a quarter of that universe is tied at zero; AUC uses the rank-based (Mann–Whitney) estimator under the midrank convention, which assigns every tied score–label pair a credit of exactly 0.5, so the reported figures are the convention-neutral expected AUC. "0" denotes an exact zero true positives in the top 500.

4.4. Direct-Defendant Scope and Bounded Reading

The direct-defendant exercise aligns the validation target with the screen's legal-economic role. Direct CADE defendants anchor the adjudicated environments but need not be persistent zero-win firms, since they may occupy winner-side, allocation, or rotating roles. The award-layer flag ranks the loser side by construction, and against direct defendants the binary flag is uninformative (AUC 0.491, random). Continuous participation volume ranks them moderately above chance (AUC 0.66–0.70): defendants are active firms, so volume carries some defendant-side information the loser-side flag does not. The data bear this out—only 14.9% of direct defendants are always-losers and their median win rate is 0.261, while the cobidder target sits in the zero-win stratum by construction. The lesson is scope, not detection: a loser-side flag is silent about the winner side of adjudicated conduct, and nothing here ranks defendants (full scope matrix in Appendix C).

Taken together, the benchmarks draw a boundary rather than certify a screen, each addressing a distinct pre-registered threat—legal anchoring, participation volume, exposure, timing, leakage, scope asymmetry (consolidated map in Appendix C). Persistent zero-win participation does concentrate adjudication-anchored loser-side exposure in the raw data, but the decomposition shows this is what procurement opportunity already implies: the residual ordering net of opportunity is marginal and fragile, the prospective ranking fails outside incumbent pools, and the raw signal sits in a few adjudicated cases. The defensible product is the audit, not the screen—a reproducible decomposition that tells an agency how far a cheap award-layer statistic can be trusted before costly bid-layer work begins (here, not far), and that transfers where the answer may differ. Section 5 runs the same audit unchanged on a second procurement system to test whether it travels; Section 6 then asks what the always-loser cobidders look like as firms.

5. The Audit on a Second Platform

Section 4 draws its boundary on a single platform. There the award-layer score concentrates adjudication-anchored loser-side exposure in the raw data, yet almost nothing survives once procurement opportunity is held fixed, and the prospective ranking collapses outside the incumbent pool. What an agency carries away is not the screen but the decomposition behind it—the sequence of checks that says how far a thin award-layer statistic can be trusted before bid-level reconstruction begins. A decomposition like that earns its place only if it survives a move to a second system. We therefore stress-test the protocol on a second platform, running it unchanged on the federal ComprasNet system, and ask what it returns where the institutions differ. The audit travels; the score does not. The protocol—and with it the first-stage opportunity deflation—reproduces on an institutionally distinct platform read against *partially overlapping* legal anchors, while the loser-side score again fails to survive the opportunity adjustment. Two cautions bound the claim. ComprasNet stress-tests the protocol's portability rather than supplying an independent cartel universe; because the two platforms share part of the same CADE record, the federal result neither confirms nor refutes any firm's legal status. And the federal within-stratum null is power-bounded: it does not by itself rule out a residual in the operationally relevant 0.55 region (Section 5.2). A genuinely independent cartel ground truth remains future work.

5.1. A Different System, the Same Protocol

ComprasNet differs from BEC–SP on exactly the dimensions that bear on the protocol, and each difference raises the bar for the test. It is a single national system aggregating procuring units across the federal administration, where BEC is one state-level platform. Its modality mix is effectively pure Pregão, so the sealed-bid Convite contrast that structures the main-text reading is absent and the construct must earn its keep with no modality margin to stratify on. And the public federal release carries participation and the winner flag but no bid microdata: an award-layer screen is the only screen one *can* build here, and the downstream bid-level reconstruction cannot be assembled from public data at all. That makes the federal panel the natural place to stress-test a thin award-layer protocol—precisely the setting an enforcement agency faces when a national platform offers participation records and little else.

We apply the Section 4 protocol unchanged. The federal panel comprises 51.0M participation rows across 92,600 firms over the 2013–2019 overlap window, of which 35,943 are always-losers. Re-estimating the same IQR rule on the federal always-loser distribution—not transporting the BEC cutoff—re-derives a federal frequent-loser threshold of 32 (against 14 on BEC) and 6,491 firms above it; the construct is the rule, not the cutoff value. The legal anchors are the *same* 7 numbered CADE adjudications, which is why the exercise reads against *partially overlapping* legal anchors rather than a fresh cartel population: the shared record lets the federal result speak to portability while keeping it from standing in for a separately assembled ground truth. Anchored cobidders number 3,850 under the broad rule and 195 under the broad always-loser restriction that mirrors the main-text target, and the frequent-loser flag never enters the label, so the federal validation is non-circular in the score by the same construction as on BEC. Federal panel construction, the CGU bulk-dump schema, and the CNPJ matching rule are detailed in Appendix G.

5.2. Side-by-Side Audit Verdict

Table 5 runs each stage of the Section 4 battery on both platforms and reports the comparable quantity in each column. The point of the table is not the absolute levels but the *shape* of the audit: whether each successive discipline—opportunity adjustment, within-stratum restriction, power-bounding, matched permutation, label-frozen timing, case concentration, negative controls—moves the federal verdict the way it moves the BEC verdict.

5.3. Interpretation

On the federal panel the protocol returns the same verdict it returns on BEC, and on the row that matters it returns it more cleanly. Both platforms over-credit the screen at the raw level: the award-layer ROC-AUC (0.744 federally) reads like discrimination until one asks what an analyst would have predicted from procurement opportunity alone. Federally the answer arrives in a single line. The exposure-only ranking, which never sees the score, reaches 0.754—matching and if anything edging past the raw score itself, where on BEC the exposure axis only approaches it. The apparent reach is opportunity arithmetic with no conduct premium left to claim.

This is not the same null encountered twice but a second coordinate on the over-crediting relationship of 1. Because the raw award score and the contact-defined label both load on participation volume, the raw AUC inflates by an amount that grows with volume dispersion and *shrinks* with the adjudicated base rate. The federal base rate runs roughly an order of magnitude below BEC's, and the characterization predicts what the federal numbers show: at the rarer target the inflation is so nearly complete that the opportunity axis alone edges *out* the raw score rather than merely approaching it. The two platforms differ in base rate and volume dispersion and land where the relationship places them. We do not estimate that relationship: it is a *synthetic surface anchored*

Table 5: The Audit on Two Platforms: BEC–SP versus ComprasNet

Audit stage	BEC–SP	ComprasNet
Raw award-layer ROC-AUC	0.761	0.744
Exposure-only ROC-AUC (no score)	0.713	0.754
Label-blind opportunity expectation (AUC)	0.553	0.611
Within-stratum residual (AUC, opportunity-decile strata)	0.471	0.462
Nested DeLong increment over exposure-only (p)	+0.010 ($p = 0.013$)	+0.005 ($p = 0.191$)
Power-bounded residual (det. prob. @ true AUC 0.55 / 0.60)[a]	0.97 / 1.00	0.35 / 0.90
Matched-permutation p (within-strata shuffle)	0.127	0.906
Label-frozen prospective timing (AUC)[b]	0.713	0.595
Top-case concentration (share of positives)	32.0%	35.4%
Negative-control verdict	generic geometry	generic opportunity/volume geometry

[a] The federal positive set is roughly 30% of BEC's, so the matched-permutation null is informative against within-stratum residuals ≥ 0.60 (detection probability 0.90) but underpowered against a residual of 0.55 (detection probability 0.35); BEC reaches 0.97 already at 0.55. The federal within-stratum positive control O_i recovers ≥ 0.99 (0.992), so the design provably detects within-stratum signal when present—the federal null is a power bound on a genuine null, not a design artifact.

[b] The federal prospective cell freezes score and always-loser pool and labels new defendant contact in 2017–2019 (positives $N^+ = 98$); the retrospective companion (0.740, $N^+ = 177$) scores in-window contact under the same frozen pool. The two are different estimands on different positive sets and are mutually consistent, not contradictory; only the prospective number is the referee's clean out-of-time test.

Notes: Each row is the comparable quantity from the Section 4 battery, re-run on the federal panel under the same definitions; the federal construction is detailed in Appendix G. Cobidder labels are adjudication-anchored exposure, not cartel membership, on both platforms; neither column ranks defendants. The federal panel has no sealed-bid modality, so no modality stratification is reported. The decisive row is the exposure-only line directly beneath raw AUC: where the opportunity axis alone matches (or, federally, edges out) the raw score, the raw number is measuring procurement opportunity, not conduct. The raw-AUC and timing rows measure different estimands on different samples—raw award-layer discrimination over the full panel versus label-frozen prospective ranking on rankable incumbents—and must not be collided. PR-AUC is the lead metric for the rare-target rows; ROC-AUC is shown for comparability, and PR-AUC levels must be read against each platform's base rate—the federal positive rate is roughly an order of magnitude lower than BEC's, so federal PR-AUC is bounded below BEC's by the rarer target, not by weaker discrimination.

at one empirical point per platform, with each platform supplying a single observed coordinate, so the two-points reading is an illustration of the mechanism's sign, not a fitted curve.

The remaining checks confirm the reading from below. Holding procurement opportunity fixed, the within-stratum residual collapses to near chance (0.462), the nested model adds nothing detectable over exposure alone (+0.005 DeLong increment, $p = 0.191$), and the matched-strata permutation does not reject ($p = 0.906$). We state the federal power bound as a bound. With a positive set about a third of BEC's, the permutation null is informative against a within-stratum residual of 0.60 but underpowered at 0.55 (0.90 versus 0.35 detection probability), so the federal deflation does not lean on the permutation alone. It rests on four power-robust pillars: the exposure-only ranking already matches the raw score, the within-stratum point estimate sits *below* 0.5, the permutation does not reject, and the negative controls return the same generic geometry. The within-stratum positive control recovers ≥ 0.99 (0.992), so the design would have caught a within-stratum signal had one been present.

One operational asymmetry deserves to be stated in the body rather than left to the supplement, because it cuts against the protocol. The federal operational signal is even more single-case concentrated than BEC's: on ComprasNet a single adjudicated case (08700.004617/2013–41) supplies 87.5% of the true positives in the top-500 forensic queue, against 45.4% for the largest BEC case. Whatever realized detections the federal ranking produces are thus overwhelmingly one cartel; the operational queue does not generalize across cases, and the cross-case claim the federal leg can support is portability of the audit, not of the ranking. (The figures are the top-case shares of top-500 true positives from the federal and BEC case-dominance outputs; Appendix G reports the full leave-one-case-out concentration battery.)

The deflation is neither a federal nor a BEC artifact; it is what a disciplined decomposition returns whenever a thin award-layer statistic is held to an opportunity-adjusted standard. An agency on either platform learns, before opening any bid record, that loser-side concentration flags opportunity and should not be carried as a stand-alone prioritization device—a lesson that binds harder federally, where the bid-level reconstruction that might follow the screen cannot be built from public data at all. Establishing the same deflation against a genuinely independent cartel ground truth, rather than the shared CADE anchors, remains future work.

5.4. Observability as an On-Thesis Finding

What each platform lets an analyst see is part of the finding, not a nuisance to control away. The federal positive base rate runs roughly an order of magnitude below BEC's, so any cross-platform PR-AUC comparison must be read against that base rate: a rarer target mechanically depresses precision-recall area, and the federal PR-AUC sits below BEC's for that reason alone, before discrimination enters. What counts as a product also differs. Federal item codes live in buyer-specific catalogs rather than a shared classification, so the item-group margin the BEC audit stratifies on has no clean federal counterpart and the federal opportunity cells are built on buyer×year margins instead (Appendix G). Those coarser cells hold opportunity less tightly fixed, which is why the federal label-blind floor (0.611) sits above BEC's (0.553)—the adjustment is conservative federally, not stronger, so the deflation read is if anything understated. And the bid tier is missing outright. Both gaps point the same way: an award-layer protocol is the protocol that survives where richer records are absent. That the construct can be re-derived, the label re-anchored, and the full deflation battery re-run on a system with no bid microdata is itself the portability result. The thin-record screen runs where bid-layer screens cannot be built, and the decomposition tells the agency how far to trust it before bid-level reconstruction—should those records ever become available—is worth commissioning.

6. Economic Content and Ordinary-Loser Alternatives

Validation establishes that the award-layer ranking concentrates adjudication-anchored loser-side exposure on the 651 always-loser cobidders, of which 341 (52.4%) satisfy the frequent-loser cut and 310 (47.6%) do not. What that target lacks so far is economic content. This section supplies it through a demanding within-stratum comparison: among always-loser firms, do those the screen targets look different from other always-losers along dimensions that would matter for loser-side prioritization? We run the contrast inside the frequent-loser stratum (341 FL cobidders against FL non-cobidders), holding the cut fixed so that no gap can be read back into it. With thousands of firms per group, p-values carry no information, so we report standardized mean differences (SMD, Cohen's d), which speak to magnitude. The full profile, the opportunity-adjusted differences, the monotonicity bins, the binary-versus-continuous comparison, and the ordinary-loser adjudication table appear in Appendix D; here we keep to the headline findings.

6.1. What opportunity adjustment removes, and what survives

The headline here cuts against the screen. Once we adjust for opportunity—the expected contact and exposure a firm would accumulate given its participation profile—the breadth, persistence, and specialization gaps between cobidders and other frequent losers *vanish*. The item-group concentration difference falls from a raw SMD of 0.13 to an adjusted -0.02, so the apparent product specialization of cobidders is a feature of the procurement environments they happen to compete in, not an intrinsic portfolio trait. The breadth gap (distinct buyers) and the persistence gap (active years) go the same way. What survives is only the mechanical defendant-contact variables: proximity to the legal anchors attenuates modestly, with the CADE-cases SMD slipping from 6.17 to 4.99. That residual is no corroboration of anything, because the label is itself built from defendant contact: the mechanical share cannot be peeled apart from a behavioral one, so the contact variables cannot stand as evidence about conduct (Appendix D).

The raw contrasts that follow are exactly the ones opportunity adjustment erases; we report them as descriptive texture, not findings. The 341 FL cobidders are predominantly electronic-auction firms—a Pregão share of 84.0% against 66.6% for the 2,394 other frequent losers (SMD 0.44). They reach more distinct buyers (21.9 versus 14.1, SMD 0.53), persist modestly longer across active years (SMD 0.39), and sit closer to the legal anchors (CADE-cases SMD 6.17)—breadth and proximity that have already failed the opportunity benchmark above.

Network specificity points the same way. Negative controls show the raw concentration is generic co-participation arithmetic rather than a signature of the adjudicated network:

swap the true CADE defendants for matched placebo anchors and the raw ranking reappears (real-anchor ROC-AUC 0.761 against a placebo-anchor mean of 0.755; $p = 0.46$), and anchors built from non-CADE high-volume winners discriminate at least as well (null mean 0.78; $p = 0.91$). Such winners sit at least as tightly inside the network as the real defendants do. The global zero-win definition does retain more raw discrimination than any market-specific alternative (best alternative 0.584 against the global 0.761), so the definition itself is not where the construct weakens (Appendix D).

The most telling diagnostic concerns what the ranking actually orders. Binning firms by participation T_i, cobidder prevalence climbs monotonically with participation—from near zero in the lowest bins to roughly 28% in the highest (rank correlation +0.99)—while opportunity-adjusted excess contact $X_i = O_i - E_i$ stays flat-to-negative across the same bins (rank correlation −0.25). Higher-ranked firms are not contacting defendants *more than their exposure predicts*; they are simply more exposed. The award-layer rank is therefore an **exposure ranking, not a collusion-intensity ranking**. This confirms the FL14 cutoff as administrative rather than structural: the continuous score $\log(1 + T_i)$ ties or beats the binary in every full-sample configuration (full-universe ROC-AUC 0.761 continuous vs. 0.688 binary), and there is no bunching at the threshold—the density ratio at $T = 14$ is 1.06, essentially flat (Appendix D). The reading is consistent with evidence that auction collusion can operate through bidder roles, repeated relationships, and cartel organization (Baldwin et al., 1997; Porter and Zona, 1999; Pesendorfer, 2000; Asker, 2010).

6.2. Ordinary-loser alternatives and what this section establishes

Could the same profile arise from the ordinary, non-collusive reasons firms lose persistently? Ordinary-loser alternatives are *narrowed but not eliminated.* Cobidders are wider and longer-lived rather than narrow and short-lived, and carry a lower Convite and minimum-quorum share, which weakens the low-capacity quorum-filler story without ruling it out. The proxies that would settle the rest—distance-to-winner, geography and logistics, later wins—are unobserved at the award layer, so benign specialization and geography stay on the table (Appendix D). The one axis these stories do not predict is the one the screen is built on: contact with the adjudicated CADE network.

On balance the limits dominate. Most of the raw distinction is procurement-environment composition and partly mechanical proximity that vanishes once opportunity is held fixed; the raw ranking is reproduced by placebo and non-CADE anchors; the rank is monotone in exposure yet flat in opportunity-adjusted excess contact; FL14 is administrative rather than structural; and ordinary-loser alternatives are narrowed, not eliminated. The construct is a disciplined exposure ranking whose non-mechanical economic content is modest. It establishes no membership, agreement, liability, or cover bidding in any tender, and on its own it is not diagnostic. Read alongside the validation of Section 4 and the sequencing of Section 7, it supports triage rather than liability: the award layer orders attention toward adjudication-anchored loser-side exposure, and richer records then judge what the screen has prioritized. Section 7 turns to implementation, asking whether the award-layer rank can be placed ahead of bid-distribution forensics in a sequence that spares bid-level reconstruction while preserving adjudication-anchored recall.

7. From Award-Layer Triage to Bid-Layer Forensics

The preceding sections validate the award-layer ranking on the margin for which it is designed. Section 4 shows that persistent zero-win participation orders adjudication-anchored loser-side exposure, and Section 6 shows that the exposed firms carry an economically distinctive profile rather than merely high procurement volume. How should an enforcement agency use such a signal? A thin administrative layer can rank priorities but cannot, on its own, evaluate the bid conduct that richer cartel evidence requires, so the useful question is one of sequencing rather than validation.

Whatever value the award-layer ranking retains lies in where it sits. It comes ahead of costly proof-producing work, built on records the agency observes broadly—who participated, who won, and who kept appearing without winning—whereas bid-layer forensics rest on submitted offers and within-tender bid moments, far more probative but far more expensive to recover, clean, and inspect at scale. The question is not whether a routine signal should replace a forensic one but whether it can govern where the forensic one is opened first. We place the award-layer ranking and an Imhof–Wallimann-style bid-distribution benchmark on the same adjudication-anchored target so that performance reads against information cost, ask whether the two layers are complements, and implement the operational gatekeeper in which award records create a survivor pool and bid forensics rank firms inside it. The object is a cost-recall frontier for forensic attention, not a horse race under full observability. This is the paper's central enforcement implication: administrative records rank where proof-producing forensics should begin, while bid-level evidence remains the stage at which conduct is evaluated.

7.1. The Evidence-Allocation Problem

Naming the decision the agency is actually solving determines what the cost-recall frontier reported below *is*. The optimal-enforcement tradition treats detection and sanctions as costly instruments chosen to maximize net deterrence (Becker, 1968; Polinsky and Shavell, 2000), and the audit literature studies when a principal pays to verify costly soft information (Townsend, 1979; Mookherjee and Png, 1989). We specialize that logic to a sequential evidence-acquisition margin: how far an agency should descend a cheap award-layer ranking before paying the costly-state-verification price of opening bid-level records. An agency with per-record forensic cost c and per-case recovery payoff V descends the ranking until the marginal recovery per forensic dollar meets the agency's cost-value ratio, $\rho(K^*)/\phi(K^*) = c/V$. This is nothing more exotic than the familiar marginal-benefit-equals-marginal-cost tangency: because the award ranking is informative—its recovery density is non-increasing in rank—the net-recovery objective is concave, so the optimal depth is the unique cost-value tangency, formalized in Appendix B (Proposition 5).

This fixes three features of the frontier. It is the *empirical image* of that cost-benefit optimum, computed on the retrospective recovery curve: each operating point is the tangency $\big(\mathrm{cost}(K^*), \mathrm{recall}(K^*)\big)$ for an agency at one c/V, and sweeping c/V traces the whole curve. One gap should not be papered over. In the model ρ is the recovery rate of prospective adjudicable cases; the curve we actually plot uses the retrospective recovery of incumbent exposure already in the CADE record, so the empirical ρ is a backward-looking proxy for the forward-looking object the proposition prices. There is no single optimal cutoff because the optimum is budget-dependent; we report the locus of optima and let an agency read off its own point. And the firm-versus-bid-row asymmetry documented below is the model's cost-side comparative static: with forensic cost indexed to bid rows and the first survivors being the highest-participation firms, marginal cost rises fastest where recovery is richest, so the firm-count reduction necessarily overstates the forensic-cost reduction.

Two boundaries are load-bearing. The stopping rule operates on recovery of the adjudication-anchored target, never on conduct: it prices the recovery footprint of an exposure-ranking architecture, ρ descends the raw exposure ranking that Section 4 has shown carries no conduct residual, and the frontier never claims that the firms opened first are likelier to have coordinated. The concavity holds on the *award-layer* recovery curve, whose density is the monotone primitive; the sequential award→bid rerank below is a second-order refinement that can dilute precision as the survivor pool grows, so realized sequential recall need not be monotone in K_1 (Section 7.3).

7.2. Bid-Layer Forensics as a Full-Observability Benchmark

The bid layer uses the submitted offers, which in the Brazilian setting are absent from the routine award file and must be recovered from the underlying `LANCES` bid histories, harmonized, and reduced to within-tender moments before any firm can be scored. It is the forensic stage, never a free comparator: it becomes available only after the agency pays to open the bid file. How much ranking information is already present before that file is opened?

The benchmark is a transparent, replicable bid-moment random forest inspired by Imhof–Wallimann-style screens rather than a tuned modern learner—a random forest trained on seven within-tender Imhof–Wallimann dispersion moments averaged across the tenders a firm contests (Abrantes-Metz et al., 2006; Imhof et al., 2018; Imhof, 2019; Huber and Imhof, 2019; Wallimann et al., 2023). The candidate pool is the always-losers for which these moments are computable, the target is the same adjudication-anchored cobidder exposure used throughout Section 4, and all 651 always-loser cobidder positives are retained. The feature mechanics, candidate-support audit, model-design audit, and leakage-boundary table appear in Appendix E.

What the layers recover.. Table 6 pairs precision, recall, and PR-AUC with ROC, since the positive base rate sits near one percent. Two facts stand out. On the pooled full-observability diagnostic the cheap award layer is *comparable* to the costly bid benchmark (award-continuous ROC 0.760 / PR-AUC 0.143; bid-layer random forest ROC 0.717 / PR-AUC 0.116)—comparable discrimination at lower informational cost, not a victory of either layer. Because the comparator is deliberately modest, a richer learner would plausibly raise the bid-layer ceiling, and that would *strengthen* the sequencing argument: a more powerful costly stage raises the value of cheap triage that decides where to deploy it. The combined model reaches ROC 0.756 / PR-AUC 0.188 but requires bid microdata for every firm, so it is a full-observability upper bound, not a first-stage rule. The pooled metrics also rest on *random* folds that are optimistic because the positives cluster by CADE case. Under case-grouped folds the bid layer falls hardest (PR-AUC $0.116 \to 0.062$, ROC $0.717 \to 0.626$), and—decisively—the combined PR-AUC falls to 0.103, *below* the award-only 0.143. The apparent complementarity does not survive case-grouped evaluation: it is the label-independent award score, not the bid layer, that keeps ranking firms when the case signal thins. Excluding the label-defining tenders leaves the bid ROC essentially unchanged ($0.717 \to 0.713$), so the fragility is case clustering, not feature–target overlap.

Do the layers see different firms?. The two layers carry partially different ranking information rather than duplicating one another: the Spearman correlation between the award and bid scores is only 0.544, and adding the award score to the bid benchmark lifts PR-AUC by +0.072 under pooled folds (DeLong $p < 0.001$). Given the case-grouped reversal, whatever marginal gain exists is mostly the award score *rescuing* the bid layer rather than the reverse. Neither score is proof of an agreement; the bid benchmark only captures the shape of a firm's bid distribution, and a strict-timing version of it is *blocked*, since within-tender dispersion moments cannot be cleanly restricted to a pre-target window, whereas the award score admits a clean temporal holdout. The full decomposition, rank-overlap diagram, and leakage boundary appear in Appendix E. The benchmark thus fixes the discriminating ceiling under this implemented full-observability comparator and shows how much of it the routine award layer already reaches at far lower informational cost.

Table 6: Bid-Layer Benchmark and Award-Layer Screen: Ranking Performance

Design	Model	PR-AUC	Prec@500	Rec@500	Prec@1000	ROC
Pooled (random folds)	Award continuous	0.143	0.216	0.166	0.177	0.760
	Award FL14	0.098	0.130	0.100	0.129	0.688
	Bid-layer random forest	0.116	0.180	0.138	0.156	0.717
	Combined award + bid	0.188	0.274	0.210	0.189	0.756
Case-grouped (held-out case)	Award continuous	0.143	0.216	0.166	0.177	0.760
	Award FL14	0.098	0.130	0.100	0.129	0.688
	Bid-layer random forest	0.062	0.086	0.066	0.083	0.626
	Combined award + bid	0.103	0.190	0.146	0.139	0.689
Excl. label-defining tenders	Bid-layer random forest	0.088	0.130	0.114	0.128	0.713

Notes: N = 16,731 always-losers with computable bid moments; 651 in-pool always-loser cobidder positives, all retained. PR-AUC is reported alongside ROC because the positive base rate is near one percent. The **pooled** block is a full-observability diagnostic with random-fold optimism; the **case-grouped** block is the honest robustness test (positives clustered by case never co-train), where the bid layer falls hardest (PR-AUC 0.116 → 0.062) and the combined model's PR-AUC (0.103) falls *below* award-only (0.143). The award scores are label-independent and so do not change across designs. Cobidders are adjudication-anchored exposure, not cartel membership; false positives are unlabeled firms, not "bad firms." The combined model is a full-observability upper bound, not a first-stage screen, and its complementarity is conditional on this implemented benchmark and case-fragile. Candidate support, model audit, complementarity, and the leakage boundary are in Appendix E.

7.3. Sequential Gatekeeping and the Cost-Recall Frontier

Similar discrimination at lower cost does not by itself establish a useful two-stage architecture; the ordering of the two layers is the operational lever. A model that hands the classifier both layers for every firm—the *joint* model—is a hypothetical full-observability ceiling that has already paid the full cost of opening the bid file, never an operating policy. The implementable object is sequential. Firms are ranked using only the award layer; the top K_1 form a survivor pool; bid microdata are recovered *only* for those survivors; and the bid benchmark then reranks them by the combined score to produce the final top-k forensic queue. Award-only is the zero-bid-cost baseline at one extreme, joint full observability the upper bound at the other, and the sequential rule is the operating architecture in between.

The frontier is not a validation of the screen but an accounting object: it tells an agency what it would cost to use a cheap exposure-ranking rule as a first-stage gate after the audit has already revealed that rule's limits. Because strict timing for the bid rerank is unavailable with the current features, the exercise is an enforcement-accounting baseline conditional on the validated incumbent ranking, not a prospective deployment test. Routing forensic recovery toward high-exposure firms, where adjudication-anchored cobidders cluster by construction, is a coherent evidence-allocation policy even with no conduct residual: the frontier prices how much of the adjudication-anchored target a capacity-constrained agency can reach for a given volume of opened bid records, without certifying that the firms opened first are likelier to have coordinated.

Why firms are the wrong denominator.. The economically meaningful cost of the rule is the volume of bid records it forces open, and that is where its value is modest. Recovering the within-tender moments requires the full tender-item `LANCES` record for every tender a survivor contested, so the cost lives in opened tender-items and bid rows, not firms, and Table 7 reports all three. At the $K_1 = 2{,}000$ operating point the rule cuts the bid-row footprint by only 33%—roughly a third—and the tender-item footprint by 34%, because opening 12% of firms still opens roughly 67% of the bid rows. The headline firm reduction of 88% is therefore the *wrong* number to lead with: it overstates the true processing-burden cut by nearly threefold. This is the cost-side comparative static of the stopping rule (Section 7.1)—with cost indexed to bid rows and recovery concentrated in high-volume survivors, the firm-count and bid-row frontiers diverge by construction—and we report all three columns so the gap stays in the body, not a footnote.

A menu of optima, not a single rule.. Varying K_1 traces the frontier that Section 7.1 identified as the locus of budget-dependent optima: each point is the tangency $\rho(K^*)/\phi(K^*) = c/V$ for an implied cost-value ratio, with cheaper forensics (lower c/V) sitting deeper on the ranking. The realized sequential award→bid counts move *non*-monotonically in K_1 at $k = 500$ (Fig. 3), because the bid rerank dilutes precision as the survivor pool grows—a second-order effect of the rerank layer, not a property of the monotone award-layer gate the proposition governs. $K_1 = 1{,}000$ in fact recovers more than $K_1 = 2{,}000$ (124 versus 116, reaching 93% of the joint upper bound's true positives at a 47% bid-row reduction), and the sequential rule never exceeds the joint bound. $K_1 = 2{,}000$ is thus the tangency at *one* implied c/V, not a privileged optimum: the contribution is the frontier itself—the menu of recall-versus-scope optima—and not any one cutoff.

What the bid rerank does, and does not, add.. At a fixed survivor pool the bid layer contributes little operationally. The award gate already gathers most of the positives the final queue recovers, and the subsequent rerank merely reorders firms inside the pool rather than surfacing positives the gate missed. The architecture earns its keep on cost—preserving much of the target capture while shrinking the scope of proof-producing forensics—not by having the bid layer locate new targets.

False positives and missed positives.. Both error margins are on the table. At $K_1 = 2{,}000$, $k = 500$ the final queue holds 116 true positives and 384 unlabeled firms (precision 0.232, recall 0.178 against the 651 positives), with 543 positives unrecovered overall. Modest precision is not failure here: in forensic triage the operative property is routing a manageable queue to the next, costlier evidentiary stage, and the unlabeled firms are unverified, not exonerated. A same-size random survivor pool sits far below the award-ranked one, confirming that the signal lives in the award ranking, not in the mere act of opening K_1 firms. A screen of this kind decides nothing about liability—only, under a stated capacity constraint, which files deserve the next and more expensive form of forensic attention.

The frontier is also case-fragile. The single largest CADE case supplies 32.0% of the positives and 45.4% of the true positives in the top-500 queue, so dropping it lowers sequential recall materially. The signal is real but concentrated: a different case mix would shift the frontier, and the operating points read as setting-specific magnitudes. The estimated ranking travels poorly as a cartel score. What travels is the decomposition—label construction, opportunity adjustment, timing discipline, case-composition audit, bid-layer comparison, and cost-recall accounting.

Timing discipline.. The frontier uses the full feature window, which sits close to the adjudication target, so the reading caveat above applies throughout. A stricter check restricts

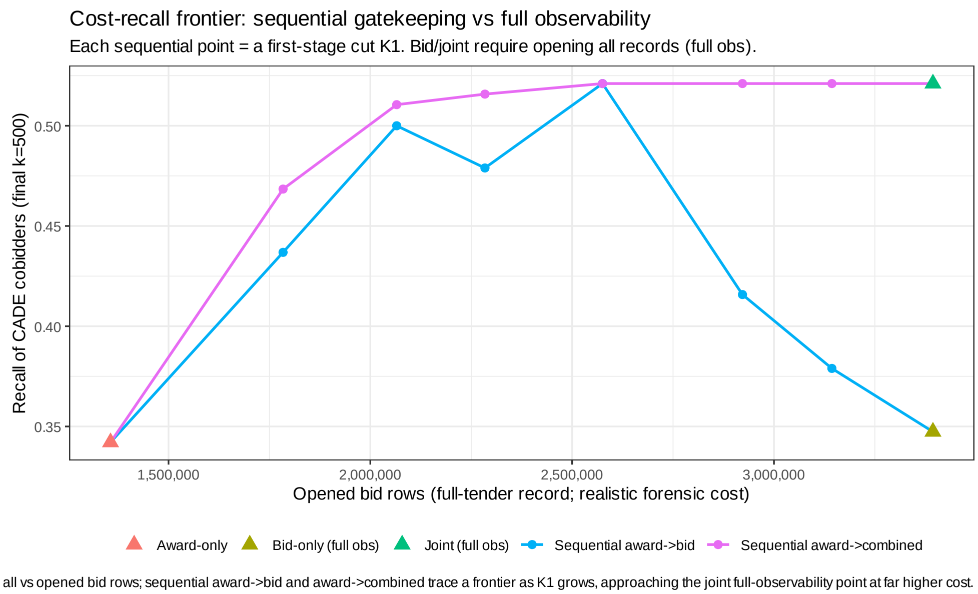

Figure 3: Cost-recall frontier for sequential award→bid gatekeeping. Points trace operational trade-offs in forensic attention, *not* legal proof. The horizontal axis is the recovered bid-record footprint (bid-row reduction relative to full observability) and the vertical axis is recall against adjudication-anchored cobidder exposure at a fixed forensic queue size. Full-observability benchmarks (bid-only, joint) require whole-pool bid recovery and are plotted as horizontal reference lines; the sequential points vary $K_1 \in \{500, 1{,}000, 2{,}000, 3{,}000, 5{,}000\}$ and never exceed the joint upper bound; award-only is the zero-bid-cost baseline. No K_1 is optimal: at $k = 500$ the recovered-positive count peaks at $K_1 = 1{,}000$.

Alt text: A scatter-and-line plot. The horizontal axis is the reduction in recovered bid records relative to opening every firm's bid history; the vertical axis is recall of adjudication-anchored cobidder positives at a fixed forensic queue size. A curve of sequential operating points rises and then falls as K_1 grows from 500 to 5,000, staying below a horizontal dashed line marking the joint full-observability upper bound; it never rises above that line. A second horizontal line marks the lower bid-only full-observability benchmark, and a marker at the far right shows the zero-bid-cost award-only baseline at lower recall. The visual message is that opening fewer bid records costs recall along a frontier, the sequential rule at best matches but never exceeds full information, and the firm-level footprint reduction overstates the bid-row footprint reduction.

Table 7: Cost-Recall Frontier for Sequential Award→Bid Gatekeeping (Forensic Queue $k = 500$)

Rule	K_1	Firms open	Tender items	Bid rows	TP	FP	Recall	Prec.	ΔFirm red.	ΔBid red.	RLoss vs joint
Award-only (no bid)	—	500	41,704	1,356,346	108	392	0.166	0.216	0.97	0.60	0.038
Bid-only (full obs)	—	16,772	116,112	3,393,915	102	398	0.157	0.204	—	—	0.048
Joint (full obs, U.B.)	—	16,772	116,112	3,393,915	133	367	0.204	0.266	—	—	0.00
Sequential	500	500	41,704	1,356,346	108	392	0.166	0.216	0.97	0.60	0.038
Sequential	1,000	1,000	58,408	1,783,750	124	376	0.190	0.248	0.94	0.47	0.014
Sequential	2,000	2,000	77,097	2,283,924	116	384	0.178	0.232	88%	33%	0.026
Sequential	3,000	3,000	87,714	2,575,274	115	385	0.177	0.230	0.82	0.24	0.028
Sequential	5,000	5,000	99,977	2,921,977	102	398	0.157	0.204	0.70	0.14	0.048

Notes: Evaluation pool: 16,772 always-loser firms with computable complete-Imhof bid-distribution features; positives are 651 always-loser cobidders. The target is *adjudication-anchored exposure*, not cartel membership. "Firms open," "Tender items," and "Bid rows" report the forensic footprint that must be recovered before the rule's final queue can be scored; full observability opens all 16,772 firms / 116,112 tender-items / 3,393,915 bid rows. **The firm count is not the only denominator**: opened tender-items and bid rows better approximate processing burden, and the firm reduction (ΔFirm red.) substantially overstates the bid-row reduction (ΔBid red.) because survivors are the highest-participation firms. **Award-only opens no bid records** as a pure award screen; the bid-footprint columns for the award-only rows are the *hypothetical* scope if it were verified at the bid layer. Bid-only and joint require full bid observability for the entire pool; the sequential rule recovers bid microdata only for the K_1 survivors. Cost reductions are computed relative to full observability. "RLoss vs joint" is recall lost relative to the joint upper bound at the same k. False positives (FP) are non-labeled firms, *not* exonerated. ROC is not the decision metric here; recall against the adjudication-anchored target at a fixed forensic queue size is. The deeper queue ($k = 1{,}000$) and the full grid (all rules, all K_1, $k \in \{100, 250, 500, 1000\}$) are in the Online Supplement; the $k = 1{,}000$ ordering is the same.

the award-layer score to a 2009–2016 window and evaluates against later adjudication-anchored exposure—not a real-time field deployment, since the labels remain adjudication-based, but a discipline on the information used to rank firms before forensics are allocated. Under that restriction the *award-only* score stays usable, at ROC 0.684 against the full-window 0.761. The *sequential* rule cannot be cleanly time-limited: the within-tender bid moments that drive the rerank cannot be restricted to a pre-target window without pulling in post-window tenders, so a strict-timing sequential frontier is *blocked*, and we say so rather than fabricate a number. The upshot is therefore retrospective and bounded. Routine administrative records can structure the order in which expensive proof-producing evidence is assembled and, on the award layer, survive a timing holdout; the richer bid layer stays the place where conduct is evaluated, and its strict timing properties cannot be certified here.

8. Scope, Limits, and Price Corroboration

Section 7 carries the paper's central enforcement-design result. Price enters only in a narrower, corroborative role: it is scope evidence on whether the award-layer screen points toward economically non-neutral environments, and nothing more. It is *not* a damages estimate, an overcharge, a cartel markup, a causal price effect of frequent-loser presence, or proof of a cover-bidding mechanism. Overcharge measurement belongs to its own tradition (Bryant and Eckard, 1991; Connor, 2007), distinct from screening (Marshall and Marx, 2012; Harrington, 2008); we draw the boundary once and hold to it. The full price grid is in Appendix F.

In the broad item-level regressions, frequent-loser presence is positively associated with log negotiated unit price ($+0.064$, $p = 0.003$; range $+3.6\% - +7.7\%$ across estimators), more strongly in pregão than in convite. That broad sign is a weighting artifact: fixing the overlap cells and moving to the within-cell average treatment effect on the treated reverses it (-0.097, $p < 0.001$), because the broad sample blends a positive between-cell selection component—flagged environments sit in structurally higher-priced cells—with a negative within-cell one. The reversal holds outside the high-value tail, the direct-CADE overlap estimate is null (-0.061, $p = 0.45$), and the within-cell movement loads on the count of genuine bidders rather than on the frequent-loser count. A selection-driven broad imprint paired with an unidentified within-cell channel keeps the evidence on the scope side of the line: the award-layer signal points to economically non-neutral environments, and that is as far as it reaches.

8.1. The Instability of a Published Administrative Cutoff

The FL14 cutoff is an administrative implementation, not a structural threshold, and Section 7 already shows the ranking carries no conduct-specific residual. The warning that matters is therefore about institutional design rather than signal robustness: any *published* administrative threshold invites manipulation once firms learn which observable patterns draw attention—the concern Sánchez Graells (2019) raises against publishing a procurement screening tool, which our setting makes concrete. Identity fragmentation and threshold-aware capping are the most corrosive responses—simulated manipulations of this kind can move the rank by several AUC points (Appendix F.6)—and a static disclosed cutoff is the easiest of all to game, because firms can bunch against it. The practical answer is to retire the cutoff, not defend it. An agency should rank firms continuously or run capacity-constrained top-k queues, refresh them as new participation data arrive, watch for bunching below any operating cutoff, and follow award-layer triage with bid-layer forensics wherever bid files can be recovered. The survivor-pool size of such a refreshed queue is itself a point on the cost–recall frontier of Section 7, so investigative capacity and exposure to gaming are chosen together rather than tuned in isolation. None of this turns the ranking into a liability rule. It is a discipline for ordering scarce investigative attention under the limits documented in Sections 7 and 8, and it should never stand as proof of conduct.

9. Conclusion

Cartel enforcement usually starts before an agency holds the evidence a case would need. Procurement records then raise a problem that precedes liability: how should scarce investigative capacity be spent when the cheap administrative record is broad but legally thin, and the richer bid-level record is costly to recover and evaluate? We treat procurement-cartel screening as evidence allocation under costly proof. The point is not to collapse the cheap award layer into the costly bid layer but to sequence them, so administrative records rank where investigation should begin and bid-level evidence judges conduct inside the prioritized set. That framing yields two analytical objects the screening literature has not written down: the enforcer's stopping rule, whose budget-dependent optimum makes the cost–recall frontier—not any single cutoff—the object to report (Proposition 5); and the over-crediting bias as an estimable quantity with a sufficient statistic a practitioner can compute before opening any bid file (Proposition 1). The deflation below is what this framework correctly catches, not a failure it confesses.

The validation evidence maps that division of labor, and its sharp limits are the substance of the finding. Persistent zero-win participation orders adjudication-anchored loser-side exposure, but a disciplined decomposition—opportunity-cell adjustment, rolling-origin timing, leave-one-case-out, and environment-dominance checks—shows that most of the cheap signal's apparent reach is procurement opportunity and case concentration. Hold participation volume and opportunity-set exposure fixed and the residual ordering turns marginal and fragile: near chance in matched cells, a small and only borderline-significant nested increment, with permutation and enrichment tests unable to reject the opportunity null, and performance riding on one adjudicated case and one procurement environment. Prospective ranking fails outside incumbent pools, since new entrants cannot be ordered by prior losing, so the ranking stays retrospective and incumbent-bound and does not behave like a generic direct-defendant detector—exactly as its loser-side scope implies. The general lesson is one of audit discipline: validating an administrative screen against adjudicated cases without adjusting for procurement opportunity systematically over-credits it. Separating this fragile residual from exposure arithmetic and case concentration, and saying plainly where each begins, is what the method contributes.

The two layers also divide labor conditionally. On the same adjudication-anchored target the award-layer score is comparable to a transparent bid-moment benchmark, yet a combined model adds information only under some validation folds and slips below the award-layer ranking under case-grouped folds. The complementarity is real but case-fragile, not a clean dominance. Joint scoring is a full-observability upper bound; sequential gatekeeping is the architecture that matters once an agency knows where the cheap layer ranks and where it does not, and the cost–recall frontier—not any single cutoff—is what it offers.

The limits are part of the design, not an apology for it. The cheap screen creates forensic priority, not proof, and the architecture travels only where award records, repeated participation, comparable procurement categories, and recoverable bid files coexist. Because a static cutoff can be gamed, use should rest on refreshed thresholds, continuous ranks or top-k queues, bunching checks, and bid-layer follow-up.

The estimated ranking is not a portable cartel score; what travels is the framework—the stopping rule, the inflation characterization, and the protocol that recovers them from label construction, opportunity adjustment, timing discipline, case-composition audit, bid-layer comparison, and cost–recall accounting. We test that transferability rather than assert it. Re-run unchanged on the federal ComprasNet platform, with partially overlapping CADE anchors and no publicly recoverable bid layer, the protocol reproduces the first-stage deflation and places the second platform where the over-crediting characterization predicts: its rarer adjudicated target is precisely why a pure opportunity ranking there edges past the raw score. The two platforms are two points on one synthetic inflation surface, not the same null twice. Cheap suspicion should be neither ignored nor mistaken for proof. Award-layer triage gives an agency a principled way to order scarce attention before the richer evidentiary record is assembled, and, just as usefully, a principled way to recognize when an administrative signal is mostly exposure and concentration rather than evidence. The statistic marks where legal attention should start, not where legal responsibility ends; charting that reach, and its edge, is what makes the cheap layer safe to work with.

Data Availability

The CADE adjudications used as legal anchors are public administrative rulings (`gov.br/cade`); the curated case file derived from them may be shared. The BEC procurement microdata—the bid-level records and the firm registry, which carry firm-identifying information (CNPJ-14) and are obtained under the terms governing access to the State of São Paulo electronic procurement platform—are administrative and confidential, and cannot be posted, redistributed, or re-licensed by the authors. We therefore intend to request the JLEO proprietary-data exemption for the BEC inputs. To allow the results to be reproduced on access-controlled inputs, we will provide the full analysis code (the Python ETL and R analysis and table/figure assembly), derived and anonymized firm-level frames (an anonymous `firm_id` replacing raw CNPJ; e.g. win-rate, always-loser flag, tenders-count, and the opportunity-adjusted firm panel), and an output-to-script map linking every reported quantity to the script that produces it. The federal ComprasNet panel used in Section 5 is built from the Comptroller-General (CGU) open bulk dumps (Portal da Transparência), restricted to the 2013–2019 window and to participation and winner-flag records only; that release carries no bid microdata.

References

Abrantes-Metz, R. M., L. M. Froeb, J. F. Geweke, and C. T. Taylor (2006). A variance screen for collusion. *International Journal of Industrial Organization 24*(3), 467–486.

Asker, J. (2010). A study of the internal organization of a bidding cartel. *American Economic Review 100*(3), 724–762.

Athey, S., J. Levin, and E. Seira (2011). Comparing open and sealed bid auctions: Evidence from timber auctions. *Quarterly Journal of Economics 126*(1), 207–257.

Bajari, P. and L. Ye (2003). Deciding between competition and collusion. *Review of Economics and Statistics 85*(4), 971–989.

Baker, J. B. (2003). The case for antitrust enforcement. *Journal of Economic Perspectives 17*(4), 27–50.

Baldwin, L. H., R. C. Marshall, and J.-F. Richard (1997). Bidder collusion at forest service timber sales. *Journal of Political Economy 105*(4), 657–699.

Becker, G. S. (1968). Crime and punishment: An economic approach. *Journal of Political Economy 76*(2), 169–217.

Bryant, P. G. and E. W. Eckard (1991). A price-fixing probability index. *Review of Economics and Statistics 73*(4), 749–752.

Chassang, S., K. Kawai, J. Nakabayashi, and J. Ortner (2022). Robust screens for noncompetitive bidding in procurement auctions. *Econometrica 90*(1), 315–346.

Conley, T. G. and F. Decarolis (2016). Detecting bidder groups in collusive auctions. *American Economic Journal: Microeconomics 8*(2), 1–38.

Connor, J. M. (2007). Price-fixing overcharges: Legal and economic evidence. In *Research in Law and Economics*, Volume 22, pp. 59–153. Elsevier.

Coviello, D. and S. Gagliarducci (2017). Tenure in office and public procurement. *American Economic Journal: Economic Policy 9*(3), 59–105.

Decarolis, F. (2014). Awarding price, contract performance, and bids screening: Evidence from procurement auctions. *American Economic Journal: Applied Economics 6*(1), 108–132.

Decarolis, F., R. Fisman, P. Pinotti, and S. Vannutelli (2025). Rules, discretion, and corruption in procurement: Evidence from italian government contracting. *Journal of Political Economy Microeconomics 3*(2), 213–254.

Decarolis, F., L. M. Giuffrida, E. Iossa, V. Mollisi, and G. Spagnolo (2020). Bureaucratic competence and procurement outcomes. *Journal of Law, Economics, and Organization 36*(3), 537–597.

DeLong, E. R., D. M. DeLong, and D. L. Clarke-Pearson (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics 44*(3), 837–845.

Fazekas, M. and G. Kocsis (2020). Uncovering high-level corruption: Cross-national objective corruption risk indicators using public procurement data. *British Journal of Political Science 50*(1), 155–164.

Harrington, J. E. (2008). Detecting cartels. In P. Buccirossi (Ed.), *Handbook of Antitrust Economics*, pp. 213–258. MIT Press.

Hovenkamp, H. (2005). *Federal Antitrust Policy: The Law of Competition and Its Practice.* Thomson West.

Huber, M. and D. Imhof (2019). Machine learning with screens for detecting bid-rigging cartels. *International Journal of Industrial Organization 65*, 277–301.

Imhof, D. (2019). Detecting bid-rigging cartels with descriptive statistics. *Journal of Competition Law & Economics 15*(4), 427–467.

Imhof, D., Y. Karagök, and S. Rutz (2018). Screening for bid rigging—does it work? *Journal of Competition Law & Economics 14*(2), 235–261.

Kawai, K. and J. Nakabayashi (2022). Detecting large-scale collusion in procurement auctions. *Journal of Political Economy 130*(5), 1364–1411.

Kawai, K., J. Nakabayashi, J. Ortner, and S. Chassang (2023). Using bid rotation and incumbency to detect collusion: A regression discontinuity approach. *The Review of Economic Studies 90*(1), 376–403.

Marshall, R. C. and L. M. Marx (2012). *The Economics of Collusion: Cartels and Bidding Rings.* Cambridge, MA: MIT Press.

Mookherjee, D. and I. Png (1989). Optimal auditing, insurance, and redistribution. *Quarterly Journal of Economics 104*(2), 399–415.

OECD (2022). Data screening tools for competition investigations. Technical Report 284, OECD Publishing, Paris.

Pesendorfer, M. (2000). A study of collusion in first-price auctions. *Review of Economic Studies 67*(3), 381–411.

Polinsky, A. M. and S. Shavell (2000). The economic theory of public enforcement of law. *Journal of Economic Literature 38*(1), 45–76.

Porter, R. H. and J. D. Zona (1993). Detection of bid rigging in procurement auctions. *Journal of Political Economy 101*(3), 518–538.

Porter, R. H. and J. D. Zona (1999). Ohio school milk markets: An analysis of bidding. *RAND*

Journal of Economics 30(2), 263–288.

Sánchez Graells, A. (2019). "Screening for Cartels" in Public Procurement: Cheating at Solitaire to Sell Fool's Gold? *Journal of European Competition Law & Practice 10*(4), 199–211.

Seibel, R. and S. Škoda (2026). Collusion by exclusion in public procurement. *American Economic Journal: Microeconomics*. Forthcoming.

Stigler, G. J. (1970). The optimum enforcement of laws. *Journal of Political Economy 78*(3), 526–536.

Townsend, R. M. (1979). Optimal contracts and competitive markets with costly state verification. *Journal of Economic Theory 21*(2), 265–293.

Wallimann, H., D. Imhof, and M. Huber (2023). A machine learning approach for flagging incomplete bid-rigging cartels. *Computational Economics 62*(4), 1669–1720.

Wils, W. P. J. (2007). Leniency in antitrust enforcement: Theory and practice. *World Competition 30*(1), 25–64.

Wils, W. P. J. (2016). The use of leniency in EU cartel enforcement: An assessment after twenty years. *World Competition 39*(3), 327–388.